Thomas Nguyen
11888002
Professor Zhe Dang
Due 3/23/26

1.

- γ is a subsequence of α
- γ is a subsequence of β
- γ does not contain abb

Run standard LCS DP with an extra dimension $s \in \{0, 1, 2\}$ tracking progress through the forbidden pattern abb. State $s = 3$ (fully matched) is never entered.

"abb" Progress Transitions $\delta(s, c)$

s	$c = a$	$c = b$	**other**
0	1	0	0
1	1	2	1
2	1	$\times$	2

$\times$ = forbidden (would complete abb)

$dp[i][j][s]$ is the length of the longest common subsequence of $\alpha[1..i]$ and $\beta[1..j]$ with abb-progress exactly s.

$$dp[i][j][s] = \max \begin{cases} dp[i-1][j][s] \\ dp[i][j-1][s] \\ dp[i-1][j-1][s'] + 1 \quad \text{if } \alpha[i] = \beta[j] \text{ and } \delta(s', \alpha[i]) = s \text{ for some valid } s' \end{cases}$$

The answer is $\max\big(dp[][n][0],\ dp[][n][1],\ dp[][n][2]\big)$.

Complexity: (n) time and space, same as standard LCS with a constant factor of 3 for the pattern states.

---

2. Let $_{1,2}$ be DFAs for $_{1}, _{2}$ with state sets $_{1}, _{2}$. Define

$$(p,) = \max_{\substack{\alpha \text{ aeed from } p \\ \beta \text{ aeed from }}} d(\alpha, \beta)$$

so that $= (_{01}, _{02})$.
$d(\alpha, \beta)$ depends only on the current DFA states, not the full strings. So we do DP over pairs $(p,) \in _{1} \times _{2}$

:

$$(p,) = \max\Big(\max_{a} (\delta_1(p,a),), \max_{b} (p, \delta_2(,b)), \max_{c} (\delta_1(p,c), \delta_2(,c)) + 1\Big)$$

with $(p,) = 0$ if $p \in {}_1,\ \in {}_2$ (both accepted), and $-$ if no accepting state is reachable from $(p,)$. $=$ iff this DP graph has a cycle among reachable productive pairs (i.e., unbounded LCS is achievable). Otherwise evaluate the DAG DP.
Complexity: $({}_1\ {}_2\ )$ time and space.

---

3. Fix a random reference string . Define

$$(\alpha) = d(\alpha,)$$

where d denotes LCS length. To find objects similar to query a, compute (a) and retrieve stored objects with $()\ \ (a)$. By the near-triangle inequality for LCS,

$$d(\alpha,) - d(\beta,)\ \ \alpha\beta$$

so similar strings (large $d(\alpha, \beta)$) receive similar hash values. Computing $(a) = d(a,)$ takes (a) time.